

BAYESIAN ESTIMATION OF A STANDARDIZED MORTALITY RATIO WITH MISSING DEATH CERTIFICATES

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Summary: We derive a new Bayesian estimation procedure for comparing Poisson-distributed mortality rates of a particular cohort and a standard population using the standardized mortality ratio (SMR) incorporating missing death certificates. Our paper assumes the realistic case of unequal unknown reporting probabilities with only vague knowledge in the form of an informative prior distribution of one of the reporting probabilities or one of the mortality rate parameters. We find closed form estimators for use with small sample sizes and a Markov chain Monte Carlo approach for cases of moderate and large sample sizes. We consider a simulation study demonstrating the effectiveness of our method and consider a real data example.

1. Introduction

Epidemiological studies often compare mortality rates in a cohort and a standard population for a particular cause of death. A common statistic for this purpose is the standardized mortality ratio (SMR), which uses counted data. The SMR is defined as X/E , where X is the observed number of deaths in a cohort due to a cause of interest and E is the expected count due to the same death cause, determined from the number of person-years in the cohort using the mortality rate from a reference population. Typically, one assumes that the observed count of a particular cause of death is Poisson

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distributed. However, problems with a simple Poisson model can occur in determining the exact cause of death due to the loss of records such as death certificates. That is, the probability p_i that a death certificate related to cause i will be available in a study is less than one, $i = 1, \dots, C$, where C is the number of causes under study, and, typically, $C = 2$.

In recent papers Rittgen and Becker (2000) and Gupta and Wang (2004) have analyzed the problem of determining the rate of death due to a particular cause when some of the death certificates are missing. In both papers, the authors have proposed Poisson count models that are adjusted for missing death certificates and derive confidence intervals for the death rate for a particular disease. Rittgen and Becker (2000) have assumed the restriction $p_1 = p_2$, but in practice this assumption may be unrealistic. Gupta and Wang (2004) have allowed for differential death certificate availability, albeit at the cost of an over-specified model. Thus, to derive their confidence intervals, Gupta and Wang (2004) have made the strong assumption that $p_1 = ap_2$, where a is known. Such an assumption is clearly problematic, but without it, identifiability problems preclude the use of their MLE method.

In our paper we develop a Bayesian approach that incorporates prior information about p_1 and p_2 but does not assume these probabilities are equal or that the proportionality constant, a , is known. Thus, our Bayesian approach enables us to make inferences for a death rate using data with incomplete death certificate information without the restrictive assumptions found in Rittgen and Becker (2000) and Gupta and Wang (2004).

The remainder of the paper is organized as follows. In Section 2 we describe the statistical model for death-rate counts incorporating data with missing death certificates. In Section 3 we construct posteriors for the

reporting probabilities and the death rate parameters. In some cases these posteriors are of closed form; for others we use iterative simulation methods. We apply our Bayesian approach to data examined by Rittgen and Becker (2000) and Gupta and Wang (2004) in Section 4. In Section 5 we consider a simulation study in which we contrast the coverage properties of an interval based on maximum likelihood methods and our Bayesian credible interval. We conclude with a brief discussion in Section 6.

2. The statistical model

Suppose there are missing death certificates so that the total number of deaths from the disease of interest in the cohort, K , is not observable. We assume that

$$K \sim \text{Poisson}(\kappa E_1),$$

where $\kappa > 0$ and E_1 is the expected number of deaths based on the number of person-years in the cohort and reference population mortality rates. Let L be the number of deaths observed over the sample of person-years not due to the disease of interest with expected number of deaths, E_2 . We assume that

$$L \sim \text{Poisson}[\lambda E_2],$$

where $\lambda > 0$. Let $Z = K + L$ be the total number of deaths in the cohort, which is observable. Then,

$$Z \sim \text{Poisson}[(\kappa + \lambda)(E_1 + E_2)],$$

where $\kappa, \lambda > 0$. Let p_i be the probability that a death certificate for cause i is available, $i = c, n$ for cancer and non-cancer respectively. Because p_1 and p_2 are unknown, we assign probability distributions to them in our Bayesian model. These prior distributions represent our uncertainty concerning their

value. Here, we assume that the joint prior distribution for (p_1, p_2) is independent of the Poisson rates.

For each subject in the study, one records whether a death certificate is available for cause 1, cause 2, or not at all. Let $X_j = 1$ if a certificate for cause 1 is available, and let $X_j = 0$, otherwise, for $j = 1, \dots, K$. Thus, for the j^{th} subject, we assume, $X_j \sim \text{Bernoulli}(p_1)$, where $0 < p_1 < 1$, and the observed number of deaths due to cause 1 is $M \equiv \sum_{i=1}^K X_i$, where $M \sim \text{Poisson}(\kappa p_1 E_1)$.

Similarly, let $Y_r = 1$ if a certificate for cause 2 is available, and let $Y_r = 0$, otherwise, for $r = 1, \dots, L$. Thus, for the r^{th} subject, we assume $Y_r \sim \text{Bernoulli}(p_2)$, where $0 < p_2 < 1$, and the observed number of deaths due to the second cause is $N \equiv \sum_{r=1}^L Y_r$, where $N \sim \text{Poisson}(\lambda p_2 E_2)$. Also, we assume X_j is independent of Y_r , for all $j \neq r$.

Here, we consider the realistic case of unknown reporting parameters p_1 and p_2 , with $p_1 \neq p_2$. To our knowledge this scenario has not been studied in the literature. For $p_1 \neq p_2$, the joint distribution for total deaths is

$$\frac{p_1^m (\kappa E_1)^m}{m!} \frac{p_2^n (\lambda E_2)^n}{n!} e^{-(\kappa E_1 + \lambda E_2)} \frac{[\kappa E_1 (1 - p_1) + \lambda E_2 (1 - p_2)]^{2-m-n}}{(z - m - n)!} \quad (1)$$

For convenience we denote the data by the column vector $\mathbf{d} = (m, n, z)'$.

3. Bayesian inference

As noted in Section 1, both Rittgen and Becker (2000) and Gupta and Wang (2004) have made strong assumptions concerning p_1 and p_2 , the former assuming that $p_1 = p_2$ and the latter assuming that $p_1 = ap_2$, with a known. The latter assumption avoids identifiability problems. This situation is similar to non-identifiability problems that arise under certain conditions in

binomial and Poisson models with misclassification. See, for instance, Joseph, Gyorkos and Coupal (1995) and Stamey, Bratcher and Young (2004).

A Bayesian analysis obviates the need to assume known values for parameters such as a , provided some prior information concerning the parameters is available. Gustafson (2005) has presented an excellent overview of this approach to handling parameter identifiability problems. Gustafson (2005) has demonstrated that with reasonably good prior information unidentified models often outperform identifiable models due to the latter's inherent bias. A Bayesian approach to inference using model (1) requires the specification of prior distributions for κ , λ , p_1 and p_2 . The data $\mathbf{d}$ allows estimation of only three of these parameters. Thus, to achieve identifiability we require an informative prior for at least one of the model parameters.

We first derive our closed-form posterior distributions for κ , λ , p_1 , and p_2 . For Poisson data the conjugate gamma prior provides a flexible family of distributions to summarize prior information. Thus, for the Poisson rates, we assume

$$\begin{aligned} & \kappa \sim \text{gamma}(\alpha_1, \beta_1) \\ \text{and} \quad & \lambda \sim \text{gamma}(\alpha_2, \beta_2). \end{aligned} \tag{2}$$

For the case of unequal death certificate availability probabilities, we assume independent beta priors:

$$\begin{aligned} & p_1 \sim \text{beta}(a_1, b_1) \\ \text{and} \quad & p_2 \sim \text{beta}(a_2, b_2). \end{aligned}$$

To obtain closed forms for the posterior distributions, we use the expansion

$$[\kappa E_1 (1-p_1) + \lambda E_2 (1-p_2)]^{z-m-n} = \sum_{j=0}^{z-m-n} \binom{z-m-n}{j} (\kappa E_1 (1-p_1))^j (\lambda E_2 (1-p_2))^{z-m-n-j}.$$

The marginal distribution can be expressed as

$$\begin{aligned} p(\mathbf{d}) &= \int_0^\infty \int_0^\infty \int_0^1 \int_0^1 p(m, n, z | \kappa, \lambda, p_1, p_2) p(\kappa) p(\lambda) p(p_1) p(p_2) dp_1 dp_2 d\kappa d\lambda \\ &= \sum_{j=0}^{z-m-n} \frac{\beta_1^{\alpha_1} \beta_2^{\alpha_2} E_1^{m+j} E_2^{z-m-j}}{\Gamma(\alpha_1) \Gamma(\alpha_2) m! n! (z-m-n)!} \\ &\quad \int_0^1 p_1^{m+\alpha_1-1} (1-p_1)^{j+b_1-1} \int_0^1 p_2^{n+\alpha_2-1} (1-p_2)^{z-m-n-j+b_2-1} \\ &\quad \times \int_0^\infty \kappa^{m+j+\alpha_1-1} e^{-\kappa(\beta_1+E_1)} \int_0^\infty \lambda^{z-m-j+\alpha_2-1} e^{-\lambda(\beta_2+E_2)} \\ &\quad \times dp_1 dp_2 d\kappa d\lambda \\ &= \sum_{j=0}^{z-m-n} \binom{m+j+\alpha_1-1}{m} \left(\frac{E_1}{\beta_1+E_1} \right)^{m+j} \left(\frac{\beta_1}{\beta_1+E_1} \right)^{\alpha_1} \\ &\quad \times \binom{z-m-j+\alpha_2-1}{z-m-j} \left(\frac{E_2}{\beta_2+E_2} \right)^{z-m-j} \left(\frac{\beta_2}{\beta_2+E_2} \right)^{\alpha_2} \\ &\quad \times \frac{B(m+a_1, j+b_1)}{B(a_1, b_1)} \frac{B(n+a_2, z-m-n-j+b_2)}{B(a_2, b_2)}. \end{aligned}$$

Through similar integration methods, one can derive the posterior densities for the four parameters, which are

$$\begin{aligned}
p(\kappa|\mathbf{d}) &= \int_0^\infty \int_0^1 \int_0^1 p(m, n, z|\kappa, \lambda, p_1, p_2) p(\nu) p(\lambda) p(p_1) p(p_2) dp_1 dp_2 d\lambda / p(\mathbf{d}) \\
&= \sum_{j=0}^{z-m-n} \binom{m+j+\alpha_1-1}{m} \left(\frac{E_1}{\beta_1+E_1} \right)^{m+j} \left(\frac{\beta_1}{\beta_1+E_1} \right)^{\alpha_1} \\
&\quad \left(\frac{z-m-j+\alpha_2-1}{z-m-j} \right) \left(\frac{E_2}{\beta_2+E_2} \right)^{z-m-j} \left(\frac{\beta_2}{\beta_2+E_2} \right)^{\alpha_2} \\
&\quad \times \frac{B(m+a_1, j+b_1)}{B(a_1, b_1)} \frac{B(n+a_2, z-m-n-j+b_2)}{B(a_2, b_2)} \\
&\quad \frac{(\beta_1+E_1)^{m+j+\alpha_1}}{\Gamma(m+j+\alpha_1)} \kappa^{m+j+\alpha_1-1} e^{-\kappa(\beta_1+E_1)} / p(\mathbf{d}), \\
p(p_1|\mathbf{d}) &= \int_0^\infty \int_0^\infty \int_0^1 p(m, n, z|\kappa, \lambda, p_1, p_2) p(\nu) p(\lambda) p(p_1) p(p_2) dp_1 d\kappa d\lambda / p(\mathbf{d}) \\
&= \sum_{j=0}^{z-m-n} \binom{m+j+\alpha_1-1}{m} \left(\frac{E_1}{\beta_1+E_1} \right)^{m+j} \left(\frac{\beta_1}{\beta_1+E_1} \right)^{\alpha_1} \\
&\quad \left(\frac{z-m-j+\alpha_2-1}{z-m-j} \right) \left(\frac{E_2}{\beta_2+E_2} \right)^{z-m-j} \left(\frac{\beta_2}{\beta_2+E_2} \right)^{\alpha_2} \\
&\quad \times \frac{B(m+a_1, j+b_1)}{B(a_1, b_1)} \frac{B(n+a_2, z-m-n-j+b_2)}{B(a_2, b_2)} \\
&\quad \frac{p_1^{m+a_1-1} (1-p_1)^{j+b_1-1}}{B(m+a_1, j+b_1)} / p(\mathbf{d}),
\end{aligned}$$

and

$$\begin{aligned}
p(p_2|\mathbf{d}) &= \int_0^\infty \int_0^\infty \int_0^1 p(m, n, z | \kappa, \lambda, p_1, p_2) p(\nu) p(\lambda) p(p_1) p(p_2) dp_2 d\kappa d\lambda / p(\mathbf{d}) \\
&= \sum_{j=0}^{z-m-n} \binom{m+j+\alpha_1-1}{m} \left(\frac{E_1}{\beta_1+E_1} \right)^{m+j} \left(\frac{\beta_1}{\beta_1+E_1} \right)^{\alpha_1} \\
&\quad \binom{z-m-j+\alpha_2-1}{z-m-j} \left(\frac{E_2}{\beta_2+E_2} \right)^{z-m-j} \left(\frac{\beta_2}{\beta_2+E_2} \right)^{\alpha_2} \\
&\quad \times \frac{B(m+a_1, j+b_1)}{B(a_1, b_1)} \frac{B(n+a_2, z-m-n-j+b_2)}{B(a_2, b_2)} \\
&\quad \frac{p_2^{n+a_2-1} (1-p_2)^{z-m-n-j+b_2-1}}{B(n+a_2, z-m-n-j+b_2)} / p(\mathbf{d}).
\end{aligned}$$

Due to the fact that the posteriors are sums of products of factorials, beta functions, and gamma functions, the closed-form posteriors are useful for only small to moderate-sized counts. In the case of large counts, posterior approximation via Markov chain Monte Carlo (MCMC) is more computationally efficient. Here, we detail a Gibbs sampling algorithm to approximate the posterior densities in the case of large counts.

To obtain the full conditionals, we augment the observed data $\mathbf{d} = (m, n, z)'$ with the unobserved true number of deaths, K and L , due to each cause. Using this augmented likelihood, one finds that the full conditional

distributions are

$$\begin{aligned}\kappa &| \lambda, p_1, p_2, K, L, \mathbf{d} \sim \text{gamma}(K + \alpha_1, E_1 + \beta_1), \\ \lambda &| \kappa, p_1, p_2, K, L, \mathbf{d} \sim \text{gamma}(L + \alpha_2, E_2 + \beta_2), \\ p_1 &| \kappa, \lambda, p_2, K, L, \mathbf{d} \sim \text{beta}(m + \alpha_1, K - m + b_1), \\ p_2 &| \kappa, \lambda, p_1, K, L, \mathbf{d} \sim \text{beta}(n + \alpha_2, L - n + b_2), \\ K &| \kappa, \lambda, p_1, p_2, L, \mathbf{d} \sim \text{shifted Poisson}(m + \alpha_1, K - m + b_1, m),\end{aligned}$$

and

$$L | \kappa, \lambda, p_1, p_2, K, \mathbf{d} \sim \text{shifted Poisson}(m + \alpha_1, K - m + b_1, n),$$

where $\text{shifted Poisson}(\mu, r)$ denotes a Poisson random variable with rate μ defined over the range $r, r + 1, r + 2, \dots$.

4. An example application

We now apply the Bayesian model proposed in Section 3 to the data analyzed by Rittgen and Becker (2000). The data, which is exhibited in Table 1, comes from a retrospective study on death rates in German foundries. For this example, the cause of death is recorded as either cancer or non-cancer. Death certificates were available for only 70% of decedents, making this problem particularly appropriate for the Bayesian approach in this paper.

Gupta and Wang (2004) have analyzed this same data allowing for different availability probabilities for the two types of death certificates. However, they assume one probability is a known, constant multiple of the other, which can lead to inconsistent results and may underestimate the overall uncertainty in the model. As indicated, we avoid this problem by allowing differential availability probabilities in our Bayesian approach provided at least one informative prior is available.

Table 1. Cancer and Non-cancer Counts

	Death Certificate Available	Expected Deaths	Death Certificate Not Available	Total
Cancer	831	881.3	NA	NA
Non-cancer	2065	2559.8	NA	NA
Total	2896	3441.1		3972

We assume non-informative priors on the two mortality rate parameters. Jeffreys priors are commonly used for priors that are non-informative relative to the likelihood. For an overview of such priors see, for example, Gelman, Carlin, Stern and Rubin (2005). The Jeffreys prior for the two Poisson rates, assuming prior independence, is $f(\kappa, \lambda) \propto 1/\sqrt{\kappa\lambda}$. Our assumption is that expert opinion exists for at least one of the availability proportions.

As an example, suppose that an expert believes p_1 has mode 0.77 and fifth percentile 0.55. This prior belief corresponds to a $\text{beta}(12.3, 4.3)$ distribution which has mean 0.738 and standard deviation 0.105. Use of software such as BetaBuster (which one can find at <http://www.epi.ucdavis.edu/diagnostictests/betabuster.html>) makes this prior determination relatively easy. Similarly, suppose our expert believes p_2 has mode 0.68 and fifth percentile 0.45. This information corresponds to a $\text{beta}(10.0, 5.2)$ prior that has mean 0.656 and standard deviation 0.118.

Because the counts are quite large, the closed-form formulae of Section 3 are not convenient in this case and, thus, we use MCMC to approximate the posterior distributions. Two chains of length 20,000 were run following a 5,000 iteration burn-in. All parameters had Gelman-Rubin values of approximately 1 indicating convergence and trace plots and autocorrelation plots all indicated convergence. The resulting posterior summary statistics for the Bayesian formulation are provided in Table 2. Following Rittgen and Becker (2000), we draw inferences on the disease rates via the standard

mortality ratios (SMR), where the SMR for the cancer deaths is $\frac{\kappa}{E_1} * 100$ and the SMR for the non-cancer deaths is $\frac{\lambda}{E_2} * 100$. For comparison purposes, in Table 3 we provide the estimates for the cancer SMR and the missing certificate probability found in Rittgen and Becker (2000). Compared to their results, our estimate for the cancer SMR is slightly smaller and has more variability. The greater variability in estimation is expected because, to realistically account for uncertainty in availability probabilities, our model has more parameters and is more flexible.

Table 2. Posterior summaries for Bayesian analysis with informative priors.

	Posterior mean	Posterior Standard Deviation	95% Credible Interval
SMR(Cancer)	126.1	16.44	(102.0, 164.3)
SMR(Non- cancer)	111.8	5.88	(98.0, 120.7)
p_1	0.753	0.089	(0.574, 0.911)
p_2	0.724	0.039	(0.671, 0.794)

Table 3. Summaries for the maximum likelihood analysis performed by Rittgen and Becker (2000). The two proportions are assumed equal in their analysis

	MLE	95% Confidence Intervals
SMR(Cancer)	129.3	(117.3, 142.5) (120.9, 137.8)
$p = p_1 = p_2$	0.73	(0.72, 0.74)

We next study the sensitivity of the posterior for the SMR for the cancer deaths (the primary parameter of interest) on the prior for p_1 . Leaving the

equivalent sample size of the prior ($a_1 + b_1 = 12.3 + 4.4 = 16.7$) the same, we consider priors with modes from 0.69 to 0.85. For instance, for a prior with mode 0.87, we use a $\text{beta}(13.8, 2.9)$ prior. The resulting posteriors are plotted in Figure 1. The posteriors all overlap substantially, but there are notable differences. Higher prior modes for p_1 result in posteriors for the SMR that are less variable and shifted more to the left. This is intuitive since the “gold standard” case of $p_1 = 1$ would have the least variability and would not result in any inflation of the posterior mean. As p_1 decreases, the posterior shifts to the right to account for the missing death certificates and adds uncertainty to the final estimation. Thus, prior sensitivity is actually expected in this case and is motivation for careful elicitation for the priors for the missing death certificate probabilities.

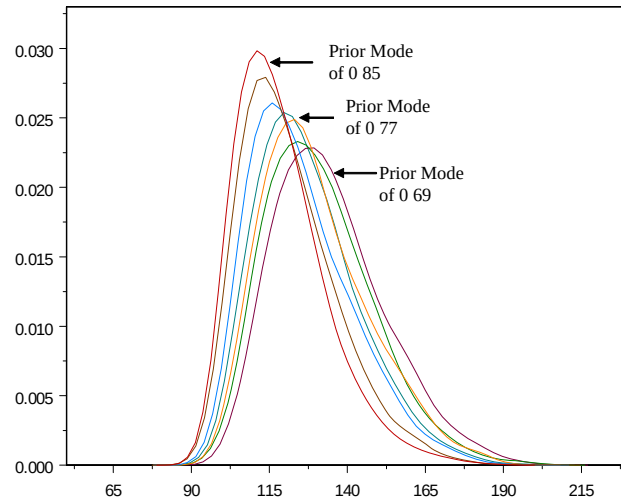


Figure 1. Posterior densities of SMR(Cancer) with prior modes ranging from 0.67 to 0.87

5. A simulation study

We now consider a Monte Carlo simulation to study the performance of the proposed Bayesian procedure. For this study we choose parameter values similar to those from the example in the previous section and assume the same missing data model as in the example. Specifically, we assume the SMR from the first population is 126, the SMR from the second population is 112, and the availability probability from the second population is 0.7. We also assume $p_1 \sim \text{beta}(32, 8)$ yielding a prior mean of 0.8 and standard deviation of 0.062. This prior distribution may seem overly informative, but in fact it is typical for Bayesian estimation methods for missing data problems; see, for example Gustafson (2005), Dendukuri, Rahme, Bélisle and Joseph (2004), among others. Indeed, in situations like this, one commonly finds priors with equivalent sample size (the sum of the beta shape parameters) as large as 100 or more.

We also include $p_1 \sim \text{beta}(20, 5)$ as a second, relatively less informative prior in our simulation. This prior has the same mean of 0.8 but has standard deviation 0.078. In this simulation we are interested in the robustness of posterior inferences to prior choices for p_1 , and, therefore, we vary the true value of the availability or reporting probability from the first population from 0.73 to 0.87. We also investigate the impact of expected count size on the performance via two different configurations of the expected counts, $E_1 = 100$, $E_2 = 200$ and $E_1 = 500$, $E_2 = 1000$. The simulation is based on 2,000 data sets and is implemented in *R*.

In this simulation we study three quantities, the relative bias, the empirical coverage of the 95% credible intervals, and the average posterior standard deviation. The relative bias is $\frac{|\hat{\theta} - \theta|}{\theta}$, where $\hat{\theta}$ represents the posterior mean and θ represents the true parameter value. Figures 2 and 3 provide

the bias and coverage probabilities for κ or, equivalently, for SMR 1. Also, Figures 4 and 5 present the bias and coverage probabilities for λ under the two different expected counts and two different priors for the reporting probability p_1 . The plots show the performance when the true value of p_1 varies from the prior mean of 0.8. As expected, the bias increases as the probability differed from the true value but never exceeded 9% for SMR 1 and was never more than 5% for SMR 2. The intervals are all conservative for values close to $p_1 = 0.8$, with coverage between 98% and 99%. For values farther away from $p_1 = 0.8$ the coverages decrease slightly below nominal but never below 94%. This simulation study shows a remarkable amount of robustness for the over-parameterized model that we considered.

This simulation indicates that the true value of p_1 has very little influence on the average posterior standard deviations. However, the prior information on p_1 and the expected counts does affect the posterior standard deviations. In an additional simulation using the same parameter values as in the previous simulation, we discovered that if one fixes the expected counts, the posterior standard deviations using a beta(32, 8) for prior 1 was between 5% and 10% smaller than when a beta(20, 5) was used for the prior across all simulation configurations. Likewise, when we fixed the prior on p_1 , the posterior standard deviations were between 30 and 35% smaller for the expected counts $E_1 = 500$, $E_2 = 1000$, than those with the expected counts $E_1 = 100$, $E_2 = 200$ across all simulation configurations. These facts are not surprising but do illustrate the impact of prior information and expected counts on posterior uncertainty.

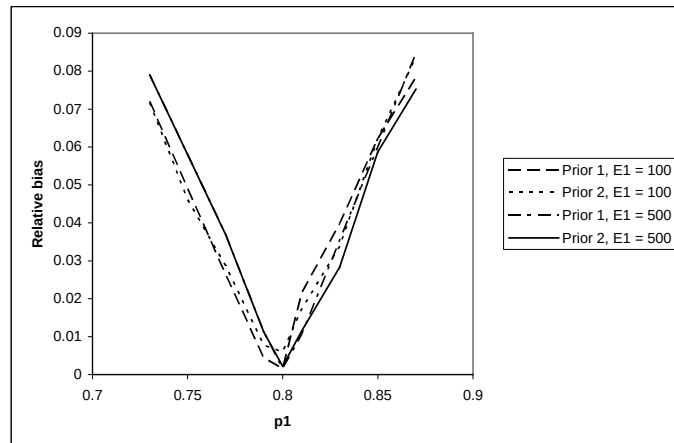


Figure 2. Relative bias for κ (SMR 1) for two priors for p_1 and two expected counts.

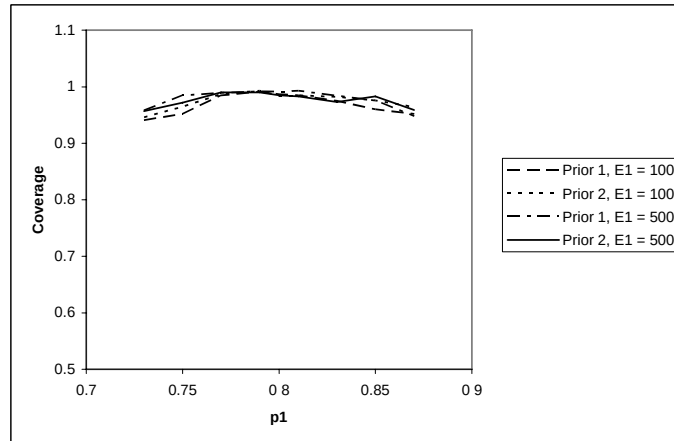


Figure 3. Coverages for 95% intervals for κ (SMR 1) for two priors for p_1 and two expected counts.

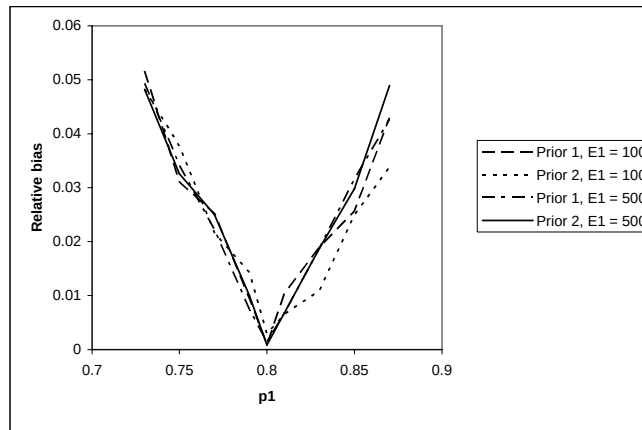


Figure 4. Relative bias for λ (SMR 2) for two priors for p_1 and two expected counts.

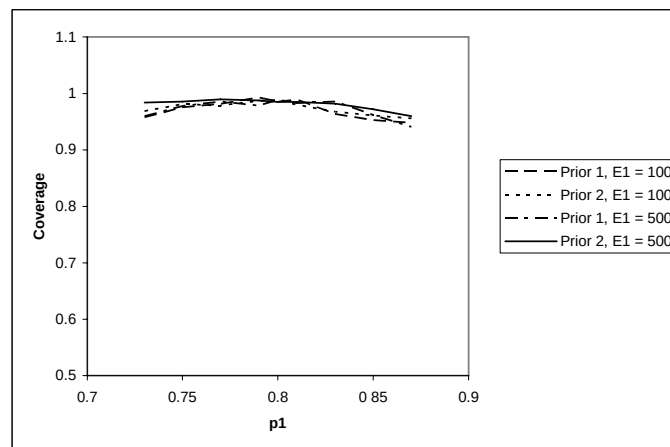


Figure 5. Coverages for 95% intervals for λ (SMR 1) for two priors for p_1 and two expected counts.

6. Summary

We have derived a new Bayesian estimation procedure for comparing Poisson-distributed mortality rates of a particular cohort and a standard population using the SMR incorporating missing death certificates. Our paper assumes the realistic case of unequal unknown reporting probabilities, $p_1 \neq p_2$. As noted in Section 5, we, like Gustafson (2005), Dendukuri, Rahme, Bélisle and Joseph (2004), and others who make this assumption, assume the availability of at least one informative prior distribution for an availability probability or one of the mortality rate parameters. By thus allowing $p_1 \neq p_2$, we generalize and improve upon the work of Rittgen and Becker (2000) and Gupta and Wang (2004), who make the strong assumptions of unknown equal reporting probabilities and known unequal reporting probabilities, respectively.

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